Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/27/2015
PI Name:	Asaithamby, Aroumougame Ph.D.		
Project Title:	Mechanism of Clustered DNA Double-strand Break Repair in Response to HZE Particles in Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7201	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/01/2013	**End Date:**	12/31/2016
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2016 per NSSC information (Ed., 10/5/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chen, David (The University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NNX13AD57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	DNA damage and consequent mutations initiate the multi-step carcinogenic process. Substantial evidence indicates that high charge and energy [HZE] particles, component of galactic cosmic rays (GCR), induced clustered DNA damages are difficult to repair, and in some instances are irreparable; this has been associated with the increased relative biological effectiveness of chromosomal aberrations, mutagenesis, and carcinogenesis. However, it is unclear why clustered lesions are difficult to repair, and the mechanisms by which unrepaired clustered DNA lesions trigger carcinogenic events are not known. The goals of this proposed research are to utilize our unique capability to quantitatively estimate the ability of cells to repair clustered DNA lesions in response to a spectrum of HZE particles, and to dissect the genetic factors that may potentially be involved in the processing of complex DNA lesions. The specific aims of the proposed research are to test the hypothesis that: (1) the cellular ability to repair HZE particles induced clustered DNA damage is modulated by radiation quality and cell cycle stages, utilizing our newly established in situ and imaging technology; (2) homologous recombination plays a key function in facilitating the repair of double-strand breaks associated with clustered DNA lesions, using key homologous recombination factors deficient cells; (3) Fanconi Anemia (FA) complex may orchestrate the repair of complex double strand breaks (DSBs) induced by HZE particles, using a panel of FA pathway deficient human cells; and (4) DNA end-processing nucleases, such as Werner Syndrome protein (WRN) and Artemis, may facilitate the processing of complex DNA ends generated by HZE particles using human cells that are either deficient in WRN and Artemis or expressing different nuclease mutant forms of WRN and Artemis. Understanding the mechanism underlying the repair of DNA damage induced by HZE particles will have important implications for estimating the risks to Astronaut health associated with GCR exposures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the mechanism underlying the repair of DNA lesions induced by HZE particles will have important implications for estimating the risks to Astronaut's health associated with GCR exposures. Our research would also lead to the development of counter-measures to enhance astronaut’s ability to sustain HZE particles exposures in space.
Task Progress:	1. Fanconi Anemia (FA) pathway factor A (FANCA) is critical for the cellular sensitivity to Fe-ions relative to gamma-irradiation. 2. FANCA is important for the repair of clustered DNA lesions induced by Fe-particle irradiation. 3. FANCA plays a role in the activation and maintenance of DNA damage response signaling following Fe-particle irradiation. 4. FANCA facilitates proper S- and G2-phase progression in response to Fe-particles irradiation. 5. FANCA stabilizes RPA2 and Rad51 at the sites of complex DSBs generated by Fe-particle irradiation. 6. FANCA protects MRE11-mediated degradation of nascent DNA-strands by stabilizing Rad51 at the site of Fe-particles induced replication stress. 7. FANCA is critical for the genome stability maintenance in response to Fe-particles exposure.
Bibliography Type:	Description: (Last Updated: 03/12/2018)
Abstracts for Journals and Proceedings	Mori E, Su F, Asaithamby A. "Multiple Roles for Rad51 in cellular response to space radiation." Presented at the 2015 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Asaithamby A, Sadek HA. "Effect of space radiation on cardiomyocyte turnover" Presented at the 2015 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Articles in Peer-reviewed Journals	Zhu J, Su F, Mukherjee S, Mori E, Hu B, Asaithamby A. "FANCD2 influences replication fork processes and genome stability in response to clustered DSBs." Cell Cycle. 2015;14(12):1809-22. Published online: 17 Jun 2015. http://dx.doi.org/10.1080/15384101.2015.1036210 ; PubMed PMID: 26083937 , Jun-2015
Articles in Peer-reviewed Journals	Gerelchuluun A, Manabe E, Ishikawa T, Sun L, Itoh K, Sakae T, Suzuki K, Hirayama R, Asaithamby A, Chen DJ, Tsuboi K. "The major DNA repair pathway after both proton and carbon-ion radiation is NHEJ, but the HR pathway is more relevant in carbon ions." Radiation Research. 2015 Mar;183(3):345-56. http://dx.doi.org/10.1667/RR13904.1 ; PubMed PMID: 25738894 , Mar-2015
Articles in Peer-reviewed Journals	Sridharan DM, Asaithamby A, Bailey SM, Costes SV, Doetsch PW, Dynan WS, Kronenberg A, Rithidech KN, Saha J, Snijders AM, Werner E, Wiese C, Cucinotta FA, Pluth JM. "Understanding cancer development processes after HZE-particle exposure: roles of ROS, DNA damage repair and inflammation." Radiation Research. 2015 Jan;183(1):1-26. http://dx.doi.org/10.1667/RR13804.1 ; PubMed PMID: 25564719 , Jan-2015
Articles in Peer-reviewed Journals	Su F, Mukherjee S, Yang Y, Mori E, Bhattacharya S, Kobayashi J, Yannone SM, Chen DJ, Asaithamby A. "Non-enzymatic role for WRN in preserving nascent DNA strands after replication stress." Cell Reports. 2014 Nov 20;9(4):1387-401. http://dx.doi.org/10.1016/j.celrep.2014.10.025 ; PubMed PMID: 25456133 , Nov-2014